Example 2.35

Find all cube roots of $\sqrt{3}+i$.

Solution

We have to find $\left(\sqrt{3}+i\right)^{\frac{1}{3}}$. Let $z=\left(\sqrt{3}+i\right)^{\frac{1}{3}}$. Then, $z^3 = \sqrt{3}+i = r\left(\cos\theta+i\sin\theta\right)$.

Then, $r = \sqrt{3+1}=2$, and $\alpha=\theta=\dfrac{\pi}{6}$ $\quad(\because \sqrt{3}+i$ lies in the first quadrant)

Therefore, $z^3 = \sqrt{3}+i = 2\left(\cos\dfrac{\pi}{6}+i\sin\dfrac{\pi}{6}\right)$

$$\Rightarrow \quad z = \sqrt[3]{2}\left(\cos\left(\frac{\pi+12k\pi}{18}\right)+i\sin\left(\frac{\pi+12k\pi}{18}\right)\right),\ k=0,1,2.$$

Taking $k=0,1,2$, we get

$k=0,$ $\quad z = 2^{\frac{1}{3}}\left(\cos\dfrac{\pi}{18}+i\sin\dfrac{\pi}{18}\right)$;

$k=1,$ $\quad z = 2^{\frac{1}{3}}\left(\cos\dfrac{13\pi}{18}+i\sin\dfrac{13\pi}{18}\right)$;

$k=2,$ $\quad z = 2^{\frac{1}{3}}\left(\cos\dfrac{25\pi}{18}+i\sin\dfrac{25\pi}{18}\right)=2^{\frac{1}{3}}\left(-\cos\dfrac{7\pi}{18}-i\sin\dfrac{7\pi}{18}\right)$.

Example 2.36

Suppose z_1, z_2, and z_3 are the vertices of an equilateral triangle inscribed in the circle $|z|=2$. If $z_1=1+i\sqrt{3}$, then find z_2 and z_3.

Solution

$|z|=2$ represents the circle with centre $(0,0)$ and radius 2.

Let *A, B,* and *C* be the vertices of the given triangle. Since the vertices z_1, z_2, and z_3 form an equilateral triangle inscribed in the circle $|z|=2$, the sides of this triangle *AB, BC,* and *CA* subtend $\dfrac{2\pi}{3}$ radians (120 degree) at the origin (circumcenter of the triangle).

(The complex number $z\,e^{i\theta}$ is a rotation of z by θ radians in the counter clockwise direction about the origin.)

Therefore, we can obtain z_2 and z_3 by the rotation of z_1 by $\dfrac{2\pi}{3}$ and $\dfrac{4\pi}{3}$ respectively.

Given that $\quad \overrightarrow{OA} = z_1 = 1+i\sqrt{3}$;

$$\overrightarrow{OB} = z_1 e^{i\frac{2\pi}{3}} = \left(1+i\sqrt{3}\right)e^{i\frac{2\pi}{3}}$$

$$= \left(1+i\sqrt{3}\right)\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$$

$$= \left(1+i\sqrt{3}\right)\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)=-2;$$

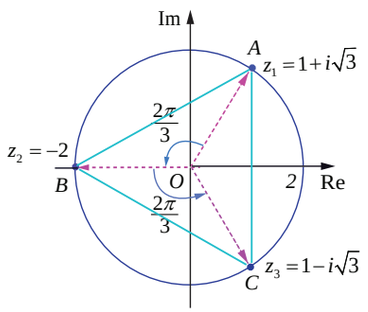


Fig. 2.48